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Experimental and Finite Element Investigations to Study the Temperature Distribution at the Orthotic Boot–Skin Interface

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Orthotic walker boots, commonly used for lower limb injuries, have been linked to discomfort due to increased skin temperatures, which can result in excessive sweating and skin problems. This research investigates the boot–skin temperature variation at the posterior and anterior sections of the leg after wearing an orthotic boot for an extended period. The temperature distribution at the orthotic boot–skin interface was studied using experimental measurements and finite element simulations. The temperature data were collected from eight male participants using 12 thermistors judiciously placed around the shank. The participants wore an orthotic boot for 60 min while sitting idle, and the temperature rises in the anterior and posterior regions of the leg were recorded. An average temperature rise of 2.3 °C ± 0.7 °C in the anterior region and 2.5 °C ± 0.6 °C in the posterior region was observed. These findings corroborate with the finite element simulations, which demonstrated similar temperature rise of 2.2 °C ± 0.4 °C and 2.4 °C ± 0.5 °C in the anterior and posterior regions, respectively. A statistical analysis using the seven-point Bedford scale for thermal sensation showed that the temperature rise in the posterior region was statistically significant ($p = 0.022$), with a higher increase noted in the posterior region compared to the anterior. The finite element simulations presented here can be used as an optimization tool to study the use of new materials and design modifications to reduce thermal discomfort in orthotic devices and exoskeletons. [DOI: 10.1115/1.4068365]

Keywords: bioheat and mass transfer, experimental/measurement techniques, biomechanics, assistive devices, prosthetics and othotics, thermal systems

Introduction

An orthotic walker boot is a passive exoskeleton commonly prescribed to patients undergoing rehabilitation following a lower leg injury (fracture, sprain) or lower leg surgery. The boot immobilizes and supports the affected lower leg, thereby unloading the ankle. Unlike traditional plaster casts, these boots can be easily donned or doffed by the patient as needed. The boots also have an air compression system that ensures a precise fit around the leg. Typically, these orthotic boots are manufactured using polyurethane foam as the cushioning material and polypropylene plastic as the supportive shell.

The authors' informal discussions with clinicians and users of the orthotic walker boots have underscored thermal discomfort as a significant concern associated with these devices which initiated investigations into this topic. To the best of the authors' knowledge, no studies have specifically discussed/studied thermal discomfort in the orthotic walker boots. However, surveys on lower limb orthoses, such as knee–ankle–foot orthoses, have reported issues like discomfort, excessive sweating, and low user satisfaction [1–3]. Orthotic walker boots and lower limb orthoses share similarities in design and construction materials, but walker boots differ in fully enclosing the lower limb with additional padding. This design restricts ventilation and skin exposure to air, which can exacerbate thermal discomfort by impeding heat dissipation at the skin–device interface. Such conditions can result in skin irritation, maceration, infections, and unpleasant odors [4]. Furthermore, recent studies indicate that patient compliance with passive orthotic devices is often poor due to thermal discomfort, improper fit, and related skin issues [5,6]. To address these challenges and enhance the orthotic device compliance, it is crucial to investigate the temperature distribution at the skin–device interface.

The problem of the thermal discomfort is not uncommon and the literature reports the investigation of thermal discomfort in other assistive devices such as exoskeletal prosthesis [7–11] and endoskeletal prosthesis [12]. Among these, thermal discomfort in lower limb prostheses has been extensively studied. While prostheses replace a missing body part and orthoses support or assist a weakened limb, these devices differ significantly in design and materials of fabrication. Despite these differences, the findings and methodologies from studies on the thermal discomfort in lower limb prostheses serve as valuable references for designing similar investigations for assistive devices like orthotic walker

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boots. Both experimental [7–11,13] and finite element (FE) simulation [14,15] approaches have been widely employed to analyze the skin–socket interface temperatures in prostheses. These methods provide a robust foundation for our study, enabling the investigation of the thermal behavior at the skin–device interface in orthotic walker boots.

The purpose of this study is to investigate the spatial and temporal variations in the boot–skin interface temperature after wearing an orthotic boot for an extended period, using both experimental methods and FE simulations. The study's outcomes will offer valuable insights into orthotics and orthotic device manufacturers, enabling them to improve the orthotic boot design by addressing the thermal discomfort experienced by the users.

Methods and Procedures

Experimental Methodology. A Tynor orthopedic walker boot of size *M* was bought over the counter for this study (see Fig. 1(*a*)). The in-house experimental setup was built to record the localized skin–boot interface temperature. Thermistors (SKU: 871999) were used to measure the temperature data. The setup comprised of 12 negative-temperature-coefficient thermistors (thermal time constant: 10 s, resistance: 10 kΩ, operating temperature range: −80 °C to 150 °C) in series with 10 kΩ resistors (tolerance: 1%, max. working voltage: 250 V). The resistor was used in series to overcome the self-heating of the thermistors, [7,8]. This circuit for 12 thermistors was combined in a printed circuit board (PCB) as shown in Fig. 1(*b*). A 2200 mAh lithium polymer battery (voltage: 11.1 V) was used as a power source. The temperature at the skin–boot interface was measured by the thermistors as resistance. It is then transferred as an analog voltage to the Arduino Mega 2560 microcontroller. The absolute measurement accuracy of thermistors was found to be ±0.2 °C using the Steinhart–Hart equation [16] in the specified operating temperature range.

The microcontroller software interface converts the analog voltage into temperature readings. The microcontroller was programmed using the ARDUINO IDE interface. The temperature data collected at the microcontroller interface were then simultaneously transferred to a data acquisition system (data streamer—Microsoft Excel) wirelessly using an HC-05 Bluetooth module. The temperature readings were then displayed on a monitor in a structured format. The detailed flowchart of the experimental data collection process is shown in Fig. 7 in Appendix.

Eight able-bodied college-going male students with a mean age of 21.3 ± 0.5 years agreed to participate in the pilot study. The participants had mean height (176.6 ± 3.9 cm), weight (72.25 ± 8.27 kg), and BMI (23.23 ± 3.13 kg/m^2). All the participants were provided written informed consent for this study participation. The study was conducted as per the revised principles of the Helsinki Declaration with minimal or no risk to study the participants. The experiment was performed in a controlled lab environment with an ambient temperature of 25 (±1.6) °C experimentally measured using thermistors held in the air for 2 min. After entering the lab, each participant was asked to rest for 5 min so the body would adjust to the ambient temperature. To measure the boot–skin interface temperature, 12 thermistors were placed on the muscular regions of the lower leg using adhesive tape and the PCB was held to the tibia using Velcro straps as shown in Fig. 1(*c*). The regions of bony prominence were avoided as the temperature variation would be much more significant in the muscular region due to metabolic activities. The locations of thermistors were divided into three sections: proximal, middle, and distal. In each section, four thermistors were placed on the sites: anteromedial, anterolateral,

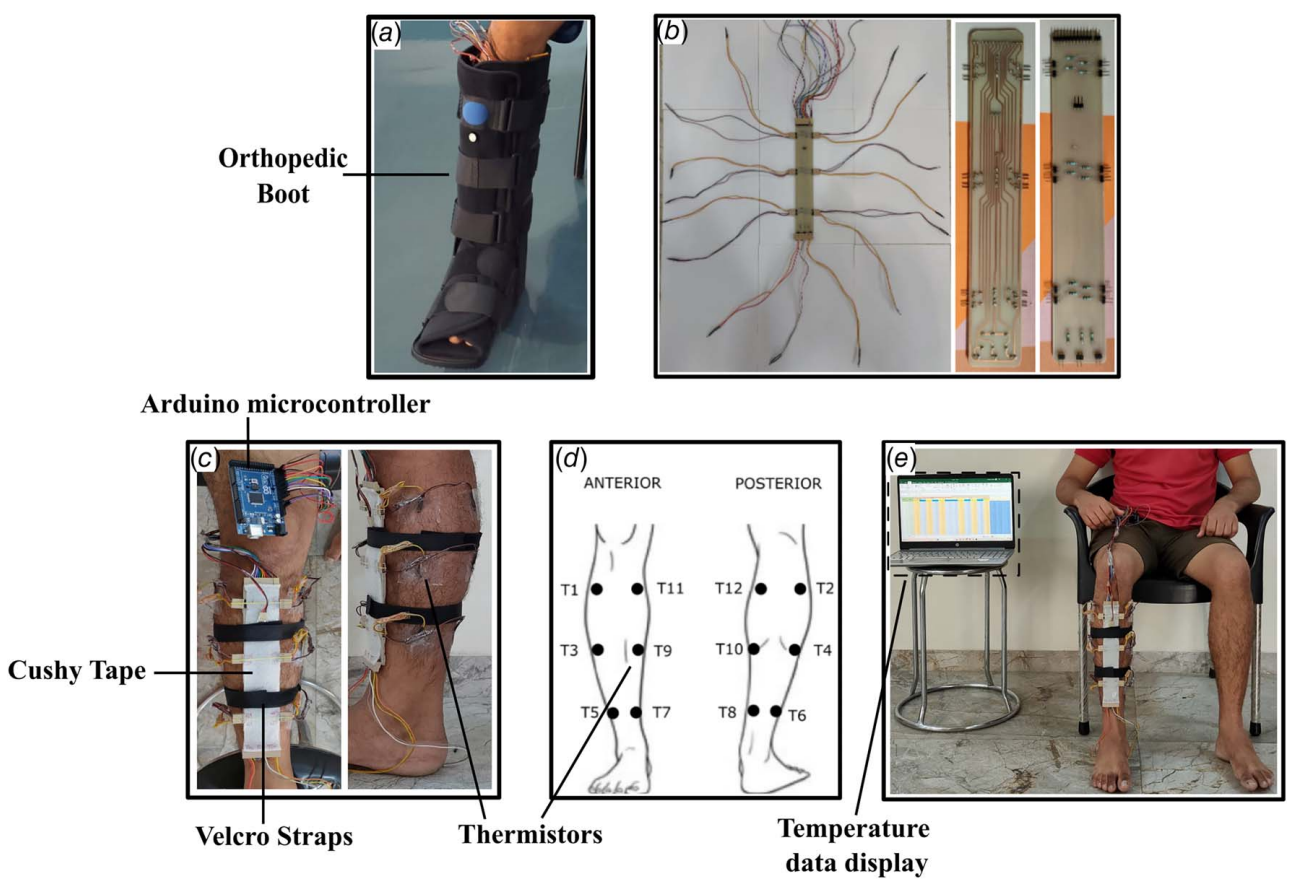

Fig. 1 Details of the experimental setup: (*a*) orthotic walker boot, (*b*) fabricated PCB and temperature sensors, (*c*) zoomed view of the temperature sensing device placed on the lower leg, (*d*) the location of thermistors on the lower leg, and (*e*) temperature sensing device attached to the subject and data being transferred to the data display

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posteromedial, and posterolateral as shown in Fig. 1(*d*), in such a way that the entire lower leg was covered (similar to Refs. [7,13]). The complete experimental setup is shown in Fig. 1(*e*), where the temperature sensing device is placed on the subject's lower limb, and the temperature data are displayed on the screen beside the subject. Temperatures were recorded for 5 min with the boot doffed to obtain the initial steady-state body temperature of the participant. The average posterior initial limb skin temperature was obtained by averaging the temperatures at thermistors T2, T4, T6, T8, T10, and T12 at the last time-step (i.e., at the fifth minute). Similarly, the average anterior initial limb skin temperature was obtained by averaging the temperatures at T1, T3, T5, T7, T9, and T11 at the last time-step (i.e., at the fifth minute). Then, the boot was donned by the participant, and any physical discomfort due to the temperature sensors was checked and carefully addressed. To prevent discomfort, a soft, cushioned tape was wrapped around the PCB (as shown in Fig. 1(*c*)), and no participant reported any issues during the study. The heat transfer study was judiciously planned with 60 min resting activity after donning the orthotic walker boot. This was the same duration as resting periods of previous experimental [8,9] and simulation [15] studies on thermal discomfort in the lower limb prostheses. The temperature readings at the skin–boot interface were recorded for 3600 s at 10-second interval. The average posterior limb skin temperature at a particular time was obtained by averaging the temperatures at thermistors T2, T4, T6, T8, T10, and T12. Similarly, the average anterior limb skin temperature was obtained by averaging the temperatures at T1, T3, T5, T7, T9, and T11.

Finite Element Simulation Methodology. A generalized finite element simulation model was built to assess the skin–boot interface temperatures after donning the orthotic boot. A simplified leg geometry was modeled using SOLIDWORKS software (Fig. 2(*a*)). The shank was modeled as a cylinder, concentrically divided into the biological tissues of bone, muscle, fat, and skin [15] as shown in Figs. 2(*b*) and 2(*c*). The foot was considered as a triangular wedge in the model. The shank and foot dimensions (shown in Fig. 2) were based on the anthropometric data available in Ref. [17]. The orthotic boot was modeled on the top of the lower leg with actual thicknesses. The computer-aided design (CAD) model assembly was imported into COMSOL MULTIPHYSICS (V6.2 AB, Stockholm, Sweden) software to perform the FE simulations. A ten-node tetrahedral element was used to mesh the CAD model, owing to complex geometry. A mesh convergence study was performed with element sizes starting from 10 mm to 3 mm. The mesh size of 7.5 mm with a total of 350,791 elements was finalized for all simulations after achieving a percentage error of less than 1% in average skin temperature. The meshed model of the leg, along with the mesh quality metrics is shown in Fig. 3. Appropriate thermal contact was ensured between the skin–boot interface using the thermal contact function in COMSOL, which evaluates the joint conductance at two contacting surfaces. Pennes' bioheat transfer equation [18] is commonly used as the governing equation to calculate the temperature distributions in tissue at basal conditions shown as follows:

$$\rho c \frac{\partial T}{\partial t} = k\nabla^2 T + Q_m + \omega_b c_b \rho_b (T - T_b) \tag{1}$$

where ρ is the density of the biological tissue (kg/m^3), c is the specific heat (J/kg/K), T is the tissue temperature (K), t is the time (s), k is the isotropic thermal conductivity (W/m/K), Q_m is the volumetric heat generation due to tissue metabolism (W/m^3), ω_b is the blood perfusion rate (1/s), c_b is the specific heat of blood, ρ_b is the density of blood, and T_b is the arterial blood temperature. In this study, heat transfer through blood perfusion was neglected for modeling simplicity. This assumption reduces Pennes' bioheat transfer

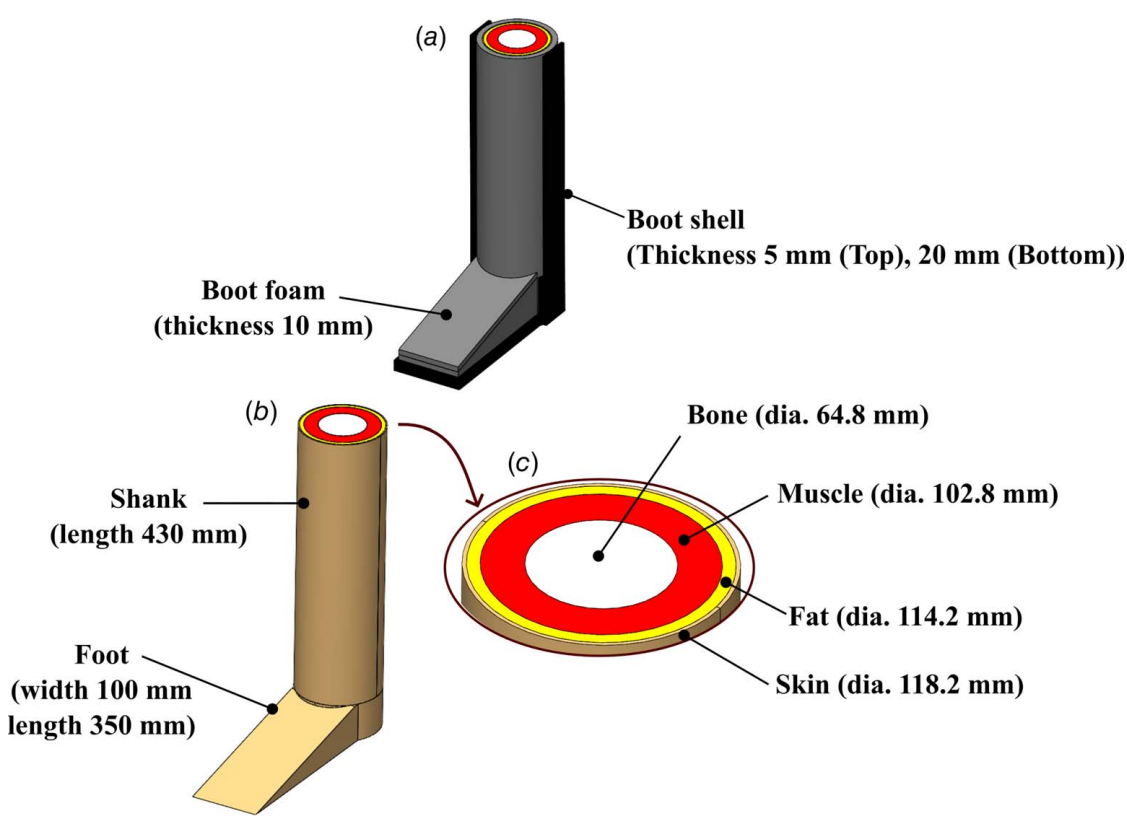

Fig. 2 Orthotic walker boot–leg CAD assembly with model dimensions: (*a*) the lower leg with the orthotic walker boot donned, (*b*) the lower leg CAD geometry, and (*c*) the enlarged view of the cross section of the leg consisting of bone, muscle, fat, and skin

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Minimum mesh size	7.5 mm
Number of Elements	350791
Element type used	Tet10
Skewness mesh quality	0.67

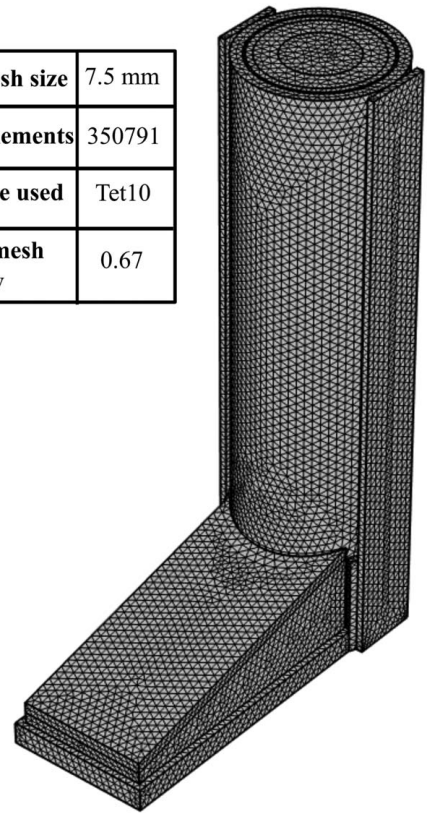

Fig. 3 Meshed CAD model of an orthotic boot–leg assembly along with the mesh metrics

equation to the heat conduction equation as follows:

$$\rho c \frac{\partial T}{\partial t} = k\nabla^2 T + Q_m \tag{2}$$

The material properties of bone, muscle, skin, fat, and orthotic boot were referred from the COMSOL material library [19] and shown in Table 1. The metabolic internal heat generation (Q_m) for each biological tissue at resting conditions was referred from Ref. [14]. The convection boundary conditions were applied on the exposed surfaces of the leg with the convection coefficient of 4 W/m^2/K [20]. The ambient temperature was set as measured in the lab. The initial anterior and posterior limb skin and foot temperatures were set according to the readings of the user without the boot measured from an experimental study. This approach helped incorporate individual baseline variations into the simulation while using the same geometric model. The initial temperatures for the biological tissues bone, muscle, and fat were set to be 36.67 °C, which is the resting core body temperature [15]. The adiabatic temperature conditions were considered at the proximal end of the leg to simplify the actual model [14,15]. With the above boundary conditions, the simulation was run for 3600 s with a time-step of 10 s. The limb skin temperatures were analyzed with time and compared with the experimental findings. The average anterior and posterior limb skin temperatures were obtained by averaging the temperatures at each element on the skin surface in that particular region.

Table 1 Material properties of the biological tissues at basal state and orthotic boot components used in the FE model [14,19]

Material	Thermal conductivity (W/m/K)	Specific heat (J/kg/K)	Density (kg/m^3)	Volumetric metabolic heat generation (W/m^3)
Bone	0.32	1313	1908	—
Muscle	0.49	3421	1090	700
Fat	0.21	2348	911	3.75
Skin	0.37	3391	1109	1120
Foam	0.035	2459	23.99	—
Polypropylene	0.141	1667	900	—

Statistical Analysis. The statistical significance of the temperature rise obtained from the experimental data in the anterior and posterior portions of the lower leg was analyzed by hypothesis testing. A seven-point Bedford scale that classifies the sensation of warmth felt by humans on a numerical scale was used to demonstrate the clinical relevance of temperature rise found in Ref. [21]. As per the Bedford scale, the absolute temperature rise is considered rather than the rate of temperature rise. According to the seven-point Bedford scale, a positive 2 refers to a temperature rise that is "too warm." This reference point was utilized to assess the significance of the temperature rise. The absolute value 2 was converted into a temperature value by using the thermal comfort model [22]. The temperature rise corresponding to "too warm" was 1.9 °C. A paired t-test was performed on the experimental temperature rise data ($p < 0.05$), with the null hypothesis indicating that the mean temperature rise of the sample is less than 1.9 °C and the alternative hypothesis suggesting that the mean temperature rise exceeds 1.9 °C, implying a significant temperature rise. Before performing the paired t-test, it was ensured that the assumptions of normality and absence of outliers were met using the Kolmogorov–Smirnov test. The statistical analysis was performed using MATLAB (R2023b, The MathWorks, Inc., Natick, MA) software.

To estimate the consistency between the FE simulations and the actual experimental measurements, the root mean square error (RMSE) was calculated for the two datasets. The RMSE calculations were performed using Microsoft Excel 2016. The RMSE was calculated using the following equation [23]:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N} (T_{i,\mathrm{Exp}} - T_{i,\mathrm{FEA}})^2}{N}} \tag{3}$$

where i denotes each time-step, and the total number of time-steps (N) is 361. $T_{i,\mathrm{Exp}}$ is the experimental temperature while $T_{i,\mathrm{FEA}}$ is the temperature (in °C) from the FE analysis at ith time-step. A time-step of 10 s over the duration of 3600 s was used.

Results

The results show that the anterior section had an average temperature rise of 2.2 ± 0.4 °C and 2.3 ± 0.7 °C while the posterior section showed an average temperature rise of 2.4 ± 0.5 °C and 2.5 ± 0.6 °C based on the FE simulations and experimental analysis, respectively. The overlay of the anterior and posterior limb skin temperatures obtained from both experimental measurements and FE simulations is shown in Figs. 4 and 5, respectively. The temporal changes in the average posterior and anterior limb skin temperatures are plotted for a duration of 3600 s. The leg temperature distribution for all participants obtained from FE simulations is shown in Fig. 6.

A paired t-test was performed to check the significance of temperature rise obtained from experimental readings for both the anterior and posterior regions. The p-values of 0.072 and 0.022 were obtained for the anterior and posterior sections, respectively, implying a significant temperature rise in the posterior region ($p < 0.05$). The results from the FE simulations were compared with the experimental results using the RMSE for each subject as shown in Table 2. The maximum and minimum root mean square errors for the anterior leg were found to be 0.84 and 0.13, respectively.

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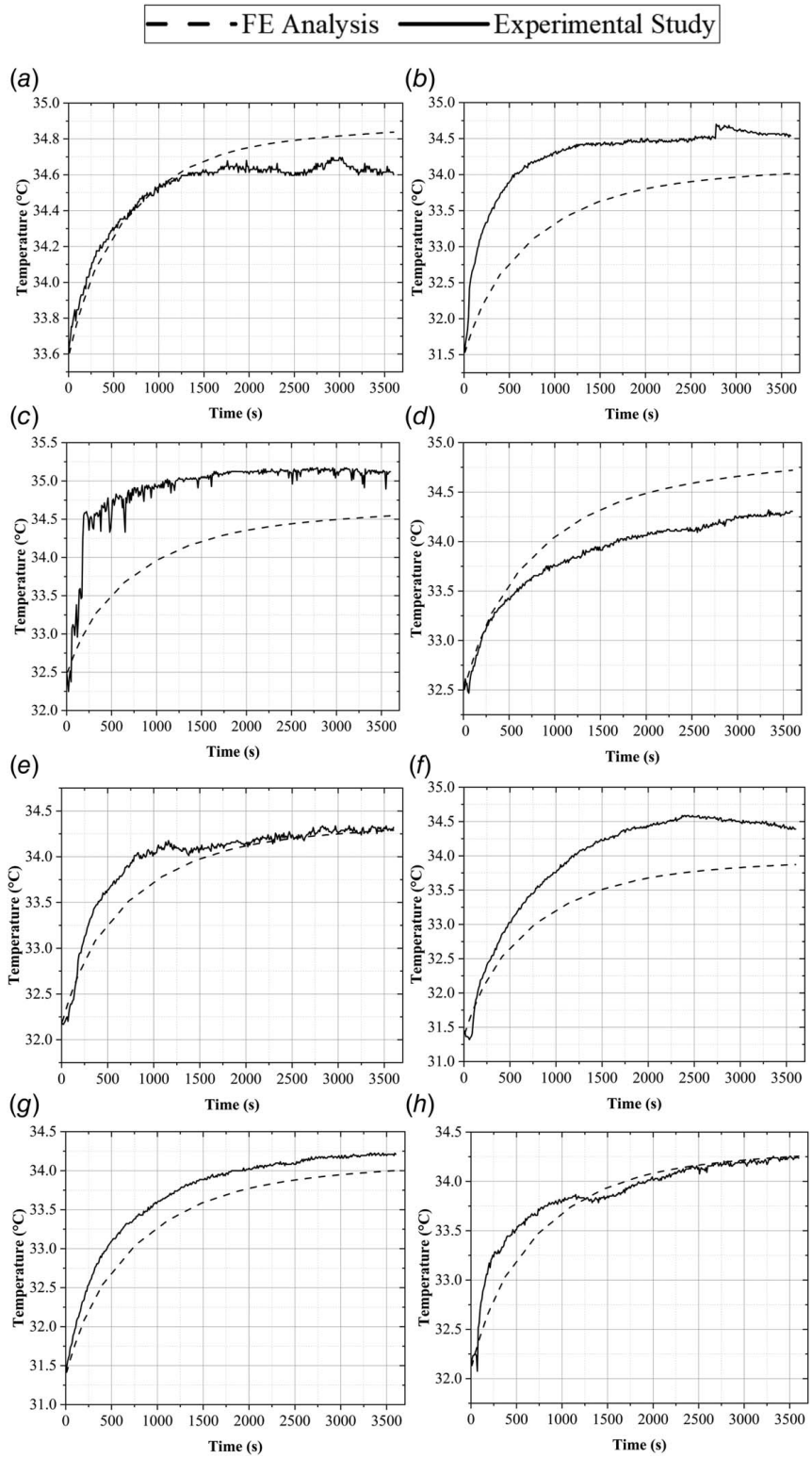

Fig. 4 Comparison of the anterior limb skin temperatures recorded from experimental and FE simulations, where (*a*)–(*h*) represent the temperature of each participant from P1 to P8 respectively

Similarly, the maximum and minimum root mean square errors for the posterior leg were 1.01 and 0.11, respectively.

Discussion

Many lower limb orthotic device users face skin-related problems and excessive sweating, eventually resulting in poor patient compliance with such devices [1,4,5]. This research aimed to investigate the temperature distribution at the orthotic boot–skin interface using the experimental and FE simulation techniques. The study's outcomes will enhance our understanding of the thermal environment at the orthotic boot–skin interface, which will help address the issues related to discomfort while wearing an orthotic boot.

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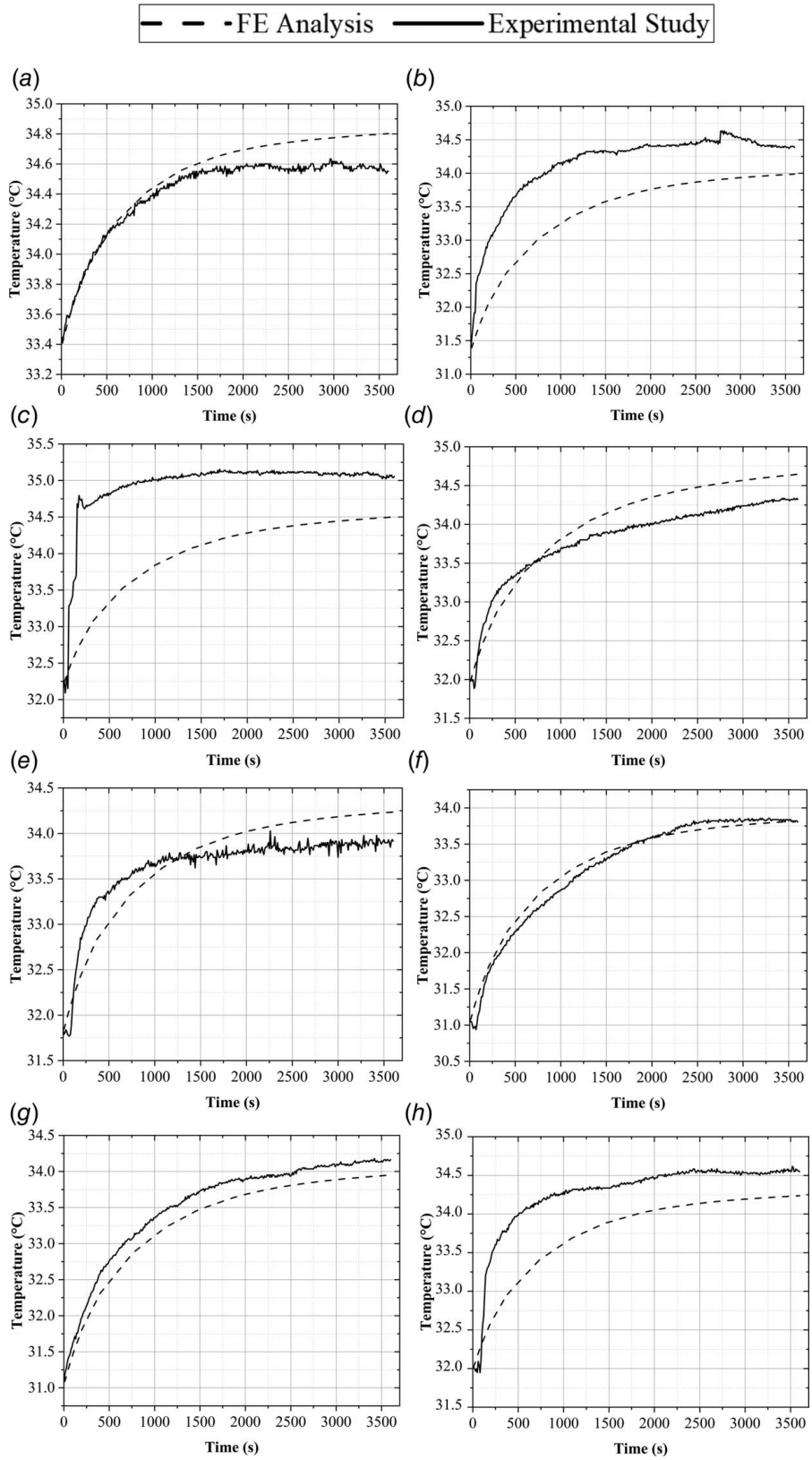

Fig. 5 Comparison of the posterior limb skin temperatures recorded from experimental and FE simulations, where (*a*)–(*h*) represent the temperature of each participant from P1 to P8 respectively

While previous studies have evaluated the biomechanical performance of orthotic walker boots [24,25], this study is the first to focus on the thermal discomfort experienced by the users of these devices. We hypothesized that wearing an orthotic walker boot for an extended period would increase the skin temperature due to restricted heat transfer caused by the boot's insulating layer. To test this hypothesis, an in-house thermistor-based temperature measurement setup was built. The interface temperature was measured

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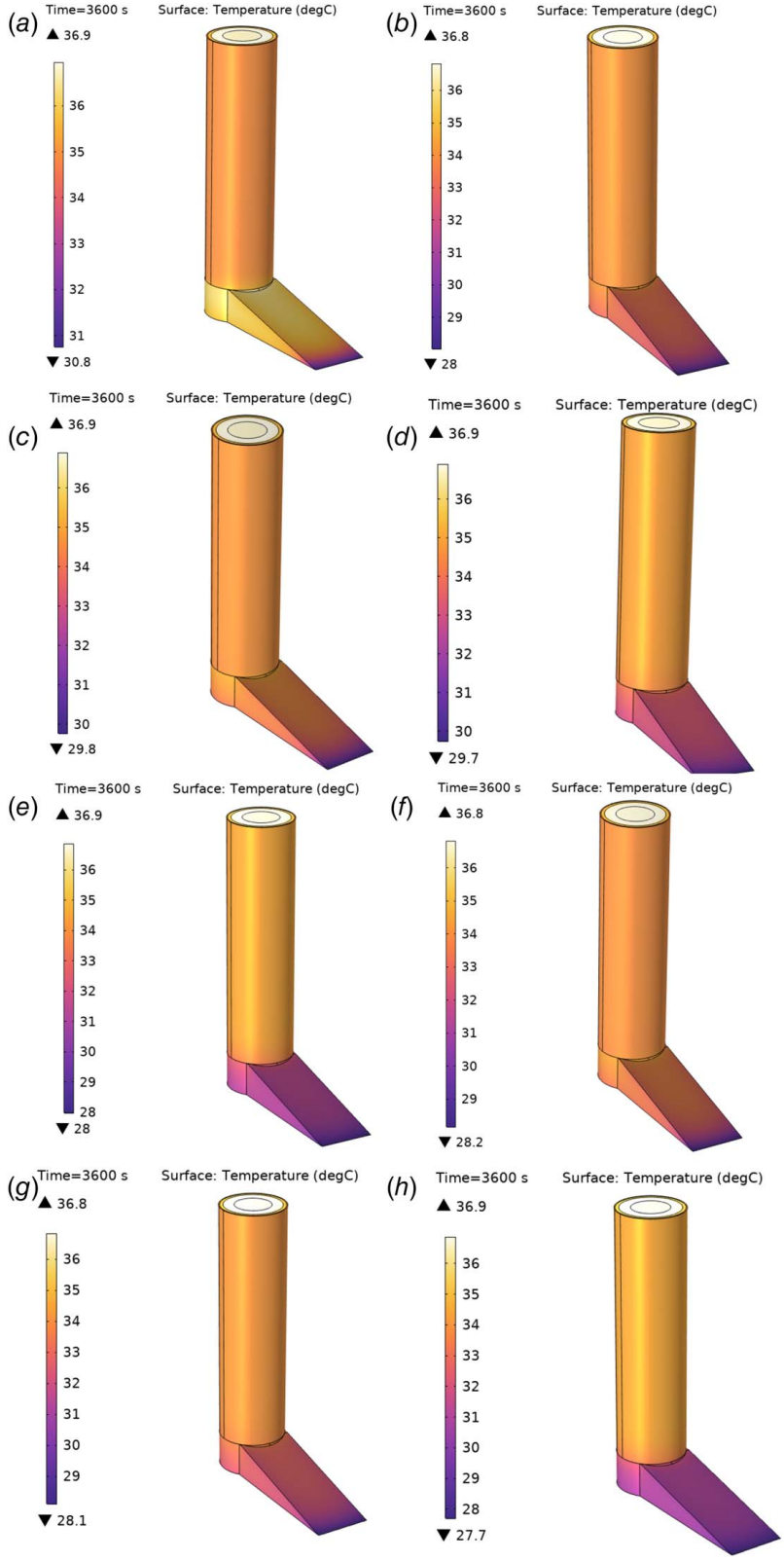

Fig. 6 Leg temperature distribution obtained from the FE results ($t = 3600$ s), where (*a*)–(*h*) represent the temperature of each subject from P1 to P8 respectively

in eight participants after they wore the orthotic boot for 60 min while sitting idle. The experimental results show a greater temperature rise (2.5 ± 0.6 °C) in the posterior leg compared to the anterior leg (2.3 ± 0.7 °C). This difference can be attributed to two factors: first, the posterior leg is covered more by the orthotic boot, which limits the heat transfer. Second, the posterior region has a higher muscular mass, leading to increased metabolic activity and, consequently, a greater temperature rise [14]. Multiple studies in the

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literature have used an experimental approach to study the skin–prosthetic socket interface temperature [7,8,13] and have reported a similar range of temperature rise. Some of these studies on the prosthetic skin–socket interfaces have reported temperature rises of 0.8–1.7 °C from donning and walking with prostheses [7] and 1.6–5.1 °C during outdoor activities [11], highlighting thermal discomfort as a significant concern for assistive device users. In the absence of studies on the thermal discomfort in orthotic devices, the findings of this study have been compared with prior research on prostheses, which served as a seminal reference for our work.

Our study is the first to apply the seven-point Bedford scale to evaluate the local thermal sensation in the lower leg, providing a novel approach to assessing the clinical relevance of temperature rise. It has been reported that a thermal sensation level of "2" corresponds to a temperature rise of 1.9 °C on the seven-point Bedford scale, implying too warm temperatures [21,22]. The paired *t*-test results showed that the temperature rise in the posterior region was statistically significant ($\Delta T > 1.9$ °C, $p = 0.022$). Peery et al. deduced from the broad literature review that a temperature rise of 1–2 °C is responsible for the thermal discomfort among assistive device wearers [7]. Thus, our study shows that orthotic boot wearers indeed experienced too warm temperatures in the posterior leg leading to thermal discomfort. These findings are not specific to the Tynor orthotic walker boot model examined in this study, as most orthotic/orthopedic walker boots available in the market use similar designs and construction materials.

Previously, the FE simulations were used to study the skin–prosthetic socket interface temperatures and the results showed that the skin temperature increased when the prosthesis was donned [14]. Another recent study evaluated the effect of the thermal conductivity of prosthetic liners on the heat dissipation inside transtibial prosthetic sockets using the FE simulations [15]. Our study also examined the skin–boot interface temperature using the FE simulations. Similar to the experimental results, the simulations show a higher temperature rise in the posterior region (2.4 ± 0.5 °C) as compared to the anterior region (2.2 ± 0.4 °C) across all participants. To evaluate the accuracy of the FE simulations, the root mean square error was calculated by comparing the FE simulation results with the experimental data. The low-RMSE values reported in Table 2 indicate a close match between the FE simulations and the experimental findings. The FE simulations presented in this study offer a valuable tool for optimizing the design of orthotic boots to reduce thermal discomfort. By exploring various combinations of lining materials, perforated designs, and the integration of active cooling systems, these simulations can guide the development of more comfortable and thermally efficient orthotic devices and exoskeletons.

In this study, a notable temperature rise was consistently observed in both the anterior (Fig. 4) and posterior (Fig. 5) parts of the leg for each participant. However, higher deviations in the experimental and FE simulation temperature data were noted for a few participants. Specifically, for the anterior region, participants P2, P3, P4, and P6, depicted in Figs. 4(*b*)–4(*d*), and 4(*f*) respectively, show considerable deviations. Similarly, for the posterior region, subjects P2, P3, and P8, shown in Figs. 5(*b*), 5(*c*), and 5(*h*) respectively, indicate similar deviations. These discrepancies are reflected in the relatively higher RMSE values for these subjects, as shown in Table 2. The FE simulations in this study used a geometrically simple and generalized CAD model of the lower leg for all eight subjects. This could have contributed to the deviations in the FE simulations owing to potential variations in the leg geometry and body fat percentage. Additionally, complex factors such as heat transfer through blood perfusion, varying metabolism rates with changes in tissue temperature, and sweating were not considered in the FE simulations. These limitations may also have contributed to deviations in the FE simulation results. In the present simulations, an adiabatic boundary condition on the proximal end of the leg was applied with reference to prior studies [14,15]. However, practically heat dissipation would occur down the leg (via blood perfusion and longitudinal conduction), and a fixed body temperature boundary condition may be more physiologically accurate. Future research should focus on developing FE simulations that utilize anatomically accurate and anthropomorphic subject-specific lower leg models, along with realistic bioheat transfer modeling approaches. For these models, the lower leg shape can be obtained from 3D scanning of the leg, while bones and soft tissues will be modeled based on the computed tomographic scan of the same subject. Based on this study's findings, we believe that a more geometrically accurate model may not significantly alter the outcomes. However, this hypothesis requires further testing and scientific validation in future research.

Table 2 The root mean square error between the experimental and FE simulation temperature measurements for each participant, from P1 to P8

Participant no.	RMSE anterior	RMSE posterior
P1	0.13	0.13
P2	0.81	0.73
P3	0.84	1.01
P4	0.36	0.28
P5	0.21	0.26
P6	0.64	0.11
P7	0.29	0.23
P8	0.17	0.54
Range (Max−Min)	0.84–0.13	1.01–0.11

As a preliminary investigation into temperature variations at the orthotic boot–skin interface, this study adopted a simplified approach by focusing on the temperature rise and stabilization in resting participants. This approach helped establish a baseline for thermal behavior without introducing the additional complexities of motion. We believe that metabolic activity and friction-induced heat generation are critical factors for thermal investigations. Future research should also investigate orthotic boot–skin temperature during dynamic activities, such as level-ground walking, walking on ramps, and stair climbing.

Since this was a pilot study, and the participant pool was limited to university students, participants were selected based on their consent and convenience. Also, since the research team was all males, females may not have approached us owing to potential awkwardness during thermistor placement. Hence, a small sample size with only male participants is a limitation of this preliminary study. Future studies with patients wearing orthotic boots need to be conducted with an adequate diversified sample size.

Conclusion

This is the first study investigating temperature at the orthotic boot–skin interface using the experimental and finite element simulation-based approaches. The statistical analysis using the Bedford scale reference showed a significant temperature rise in the posterior leg corresponding to the too warm temperature. The average anterior and posterior temperatures measured experimentally, closely matched the FE simulation results with small root mean square deviations. The FE simulations using a simplified CAD model demonstrated a close match with the experimental results, suggesting that a more complex model may not be required to achieve accurate predictions. Furthermore, this validation offers potential benefits for future design initiatives, as various combinations of materials can be explored for the orthotic walker boot optimization using FE simulations. This study provides a robust methodology for analyzing the thermal environment of wearable exoskeletons and assistive devices, offering valuable insights for enhancing user comfort, leading to device acceptance.

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Funding Data

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Appendix

Fig. 7

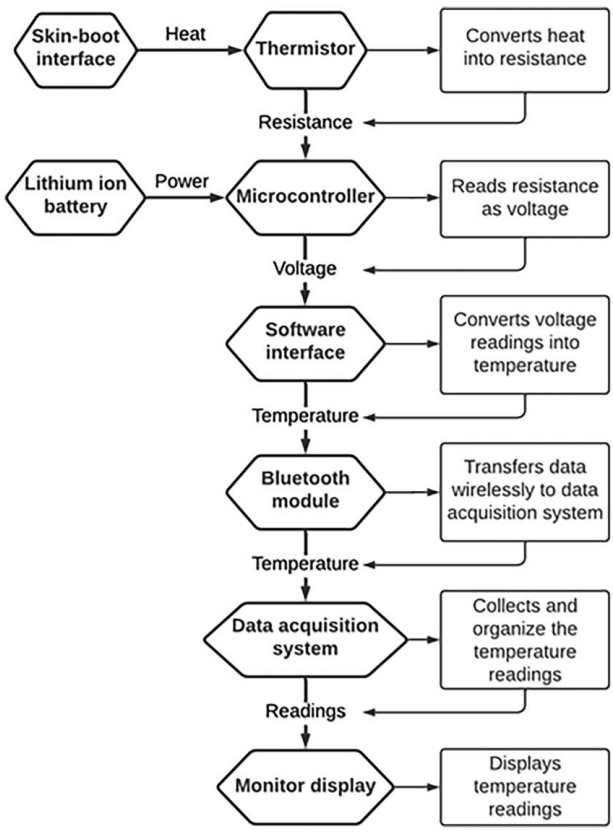

Fig. 7 Flowchart of an experimental data collection process

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